Open-Meteo

Weather Forecast API

Seamless integration of high-resolution weather models with up 16 days forecast

Location and Time

Location: Coordinates List Bounding box

Latitude
52,52

Longitude
13,41

Timezone
Not set (GMT+0)

Search

Time: Forecast Length Time Interval

Forecast days
7 days (default)

Past days
0 days (default)

By default, we provide forecasts for 7 days, but you can access forecasts for up to 16 days. If you're interested in past weather data, you can use the Past Days feature to access archived forecasts.

Hourly Weather Variables

- [x] Temperature (2 m)
- [ ] Relative Humidity (2 m)
- [ ] Dewpoint (2 m)
- [ ] Apparent Temperature
- [ ] Precipitation Probability
- [ ] Precipitation (rain + showers + snow)
- [ ] Rain
- [ ] Showers
- [ ] Snowfall
- [ ] Snow Depth
- [ ] Weather code
- [ ] Sea Level Pressure
- [ ] Surface Pressure
- [ ] Cloud Cover Total
- [ ] Cloud Cover Low
- [ ] Cloud Cover Mid
- [ ] Cloud Cover High
- [ ] Visibility
- [ ] Evapotranspiration
- [ ] Reference Evapotranspiration (ET_0)
- [ ] Vapour Pressure Deficit

- [ ] Wind Speed (10 m)
- [ ] Wind Speed (80 m)
- [ ] Wind Speed (120 m)
- [ ] Wind Speed (180 m)
- [ ] Wind Direction (10 m)
- [ ] Wind Direction (80 m)
- [ ] Wind Direction (120 m)
- [ ] Wind Direction (180 m)
- [ ] Wind Gusts (10 m)
- [ ] Temperature (80 m)
- [ ] Temperature (120 m)
- [ ] Temperature (180 m)
- [ ] Soil Temperature (0 cm)
- [ ] Soil Temperature (6 cm)
- [ ] Soil Temperature (18 cm)
- [ ] Soil Temperature (54 cm)
- [ ] Soil Moisture (0-1 cm)
- [ ] Soil Moisture (1-3 cm)
- [ ] Soil Moisture (3-9 cm)
- [ ] Soil Moisture (9-27 cm)
- [ ] Soil Moisture (27-81 cm)

Additional Variables And Options

Solar Radiation Variables

Pressure Level Variables

Weather models

15-Minutely Weather Variables

Daily Weather Variables

- [ ] Weather code
- [ ] Maximum Temperature (2 m)
- [ ] Minimum Temperature (2 m)
- [ ] Maximum Apparent Temperature (2...
- [ ] Minimum Apparent Temperature (2 ...
- [ ] Sunrise
- [ ] Sunset
- [ ] Daylight Duration
- [ ] Sunshine Duration
- [ ] UV Index
- [ ] UV Index Clear Sky
- [ ] Rain Sum
- [ ] Showers Sum
- [ ] Snowfall Sum
- [ ] Precipitation Sum
- [ ] Precipitation Hours
- [ ] Precipitation Probability Max
- [ ] Maximum Wind Speed (10 m)
- [ ] Maximum Wind Gusts (10 m)
- [ ] Dominant Wind Direction (10 m)
- [ ] Shortwave Radiation Sum
- [ ] Reference Evapotranspiration (ET_0)

Additional Daily Variables

Current Weather

- [ ] Temperature (2 m)
- [ ] Relative Humidity (2 m)
- [ ] Apparent Temperature
- [ ] Is Day or Night
- [ ] Precipitation
- [ ] Rain
- [ ] Showers
- [ ] Snowfall

- [ ] Weather code
- [ ] Cloud Cover Total
- [ ] Sea Level Pressure
- [ ] Surface Pressure
- [ ] Wind Speed (10 m)
- [ ] Wind Direction (10 m)
- [ ] Wind Gusts (10 m)

Note: Current conditions are based on 15-minutely weather model data. Every weather variable available in hourly data, is available as current condition as well.

Settings

Temperature Unit
Celsius °C

Wind Speed Unit
km/h

Precipitation Unit
Millimeter

Timeformat
ISO 8601 (e.g. 2026-04-22)

Usage licence: Non-Commercial Commercial Self-Hosted

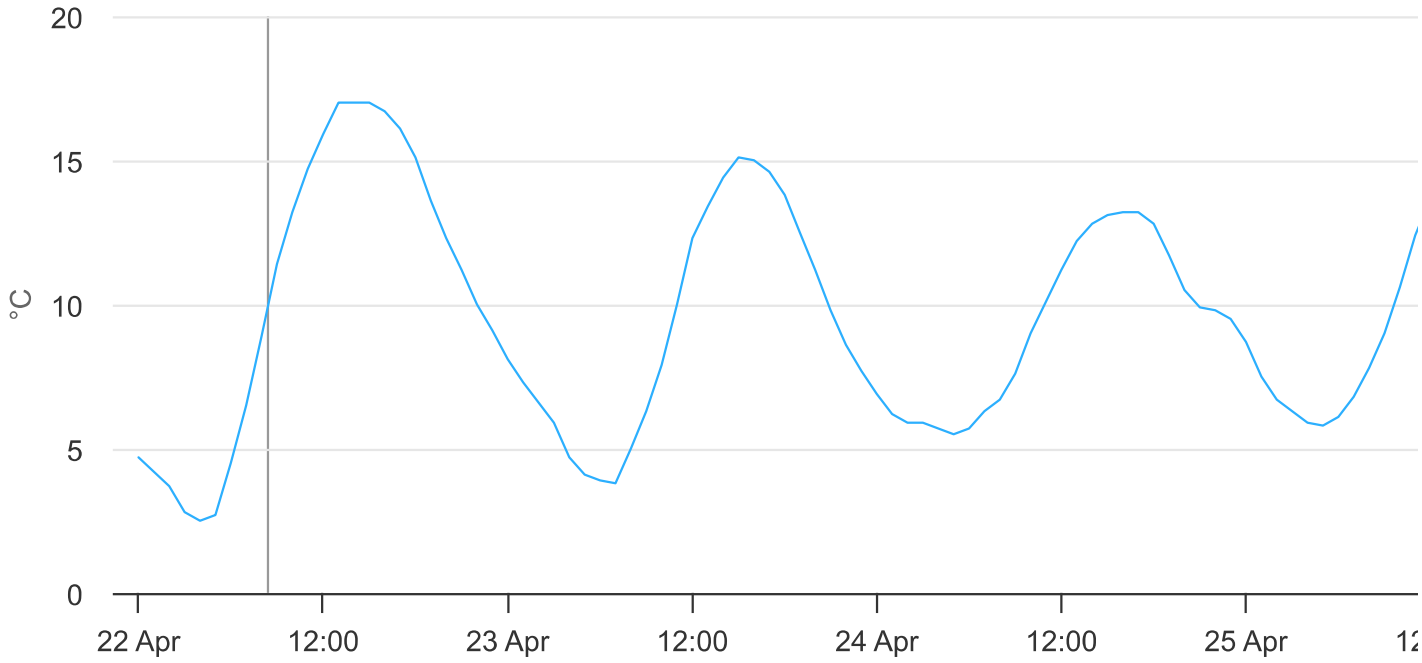
Only for **non-commercial use** and less than 10.000 daily API calls. See Terms for more details.

API Response

Preview: Chart & URL Python TypeScript Swift Other

52.52°N 13.4

Generated in 0.05ms,



Download XLSX Download CSV

API URL (Open in new tab or copy this URL into your application)

https://api.open-meteo.com/v1/forecast?latitude=52.52&longitude=13.41&hour

Data Sources

Open-Meteo weather forecast APIs use weather models from multiple national weather providers. For each location worldwide, the best models will be combined to provide the best possible forecast.

Weather models cover different geographic areas at different resolutions and provide different weather variables. Depending on the model, data have been interpolated to hourly values or not all weather variables are available. With the drop down Weather models (just below the hourly variables), you can select and compare individual weather models.

Weather Model	National Weather Provider	Origin Countr
ICON	Deutscher Wetterdienst (DWD)	Germany
GFS & HRRR	NOAA	United States

Weather Model	National Weather Provider	Origin Countr
ARPEGE & AROME	Météo-France	France
IFS & AIFS	ECMWF	European Unic
UKMO	UK Met Office	United Kingdo
KMA	KMA Korea	Korea
MSM & GSM	JMA	Japan
ICON CH	MeteoSwiss	Switzerland
MET Nordic	MET Norway	Norway
GEM	Canadian Weather Service	Canada
ACCESS-G	Australian Bureau of Meteorology (BOM)	Australia
GFS GRAPES	China Meteorological Administration (CMA)	China
HARMONIE	KNMI	Netherlands
HARMONIE	DMI	Denmark
ARPAE	ItaliaMeteo	Italy

You can find the update timings in the model updates documentation.

API Documentation

The API endpoint /v1/forecast accepts a geographical coordinate, a list of weather variables and responds with a JSON hourly weather forecast for 7 days. Time always starts at 0:00 today and contains 168 hours. If &forecast_days=16 is set, up to 16 days of forecast can be returned. All URL parameters are listed below:

Parameter	Format	Required	Default	Description
latitude, longitude	Floating point	Yes		Geographical WGS84 coor &latitude=52.52,48.85&lo

Parameter	Format	Required	Default	Description
				list of structures. CSV and negative longitudes, becau
elevation	Floating point	No		The elevation used for stat manually set the elevation disabled and the API uses separated.
hourly	String array	No		A list of weather variables parameter in the URL can
daily	String array	No		A list of daily weather varia &daily= parameter in the
current	String array	No		A list of weather variables
temperature_unit	String	No	celsius	If fahrenheit is set, all tem
wind_speed_unit	String	No	kmh	Other wind speed speed u
precipitation_unit	String	No	mm	Other precipitation amour
timeformat	String	No	iso8601	If format unixtime is selec timestamp are in GMT+0! correct date.
timezone	String	No	GMT	If timezone is set, all time time zone name from the automatically resolved to t specified.
past_days	Integer (0-92)	No	0	If past_days is set, yesterd
forecast_days	Integer (0-16)	No	7	Per default, only 7 days are
forecast_hours **forecast_minutely_15** **past_hours** **past_minutely_15**	Integer (>0)	No		Similar to forecast_days, th current day as a reference,

Parameter	Format	Required	Default	Description
start_date end_date	String (yyyy-mm-dd)	No		The time interval to get we
start_hour end_hour start_minutely_15 end_minutely_15	String (yyyy-mm-ddThh:mm)	No		The time interval to get we 2022-06-30T12:00).
models	String array	No	auto	Manually select one or mo
cell_selection	String	No	land	Set a preference how grid- to the requested coordina the nearest possible grid-c
apikey	String	No		Only required to commerc customer- . See pricing fo

Additional optional URL parameters will be added. For API stability, no required parameters will be added in the future!

Hourly Parameter Definition

The parameter &hourly= accepts the following values. Most weather variables are given as an instantaneous value for the indicated hour. Some variables like precipitation are calculated from the preceding hour as an average or sum.

Variable	Valid time	Unit	Description
temperature_2m	Instant	°C (°F)	Air temperature at 2 mete
relative_humidity_2m	Instant	%	Relative humidity at 2 met
dew_point_2m	Instant	°C (°F)	Dew point temperature at
apparent_temperature	Instant	°C (°F)	Apparent temperature is t factor, relative humidity a
pressure_msl surface_pressure	Instant	hPa	Atmospheric air pressure Typically pressure on mea

Variable	Valid time	Unit	Description
			with increasing elevation.
cloud_cover	Instant	%	Total cloud cover as an ar
cloud_cover_low	Instant	%	Low level clouds and fog
cloud_cover_mid	Instant	%	Mid level clouds from 3 tc
cloud_cover_high	Instant	%	High level clouds from 8 k
wind_speed_10m **wind_speed_80m** **wind_speed_120m** **wind_speed_180m**	Instant	km/h (mph, m/s, knots)	Wind speed at 10, 80, 120 standard level.
wind_direction_10m **wind_direction_80m** **wind_direction_120m** **wind_direction_180m**	Instant	°	Wind direction at 10, 80,
wind_gusts_10m	Preceding hour max	km/h (mph, m/s, knots)	Gusts at 10 meters above
shortwave_radiation	Preceding hour mean	W/m²	Shortwave solar radiation global horizontal irradiati
direct_radiation **direct_normal_irradiance**	Preceding hour mean	W/m²	Direct solar radiation as a normal plane (perpendicu
diffuse_radiation	Preceding hour mean	W/m²	Diffuse solar radiation as
global_tilted_irradiance	Preceding hour mean	W/m²	Total radiation received o calculation is assuming a azimuth parameter. Tilt ra should be close to 0° (0° s "nan", the calculation assu assumed that the panel h axial tracker is assumed.

Variable	Valid time	Unit	Description
vapour_pressure_deficit	Instant	kPa	Vapour Pressure Deficit (V transpiration of plants inc
cape	Instant	J/kg	Convective available pote
evapotranspiration	Preceding hour sum	mm (inch)	Evapotranspration from la location. Available soil wa liter of water per spare me
et0_fao_evapotranspiration	Preceding hour sum	mm (inch)	ET_0 Reference Evapotrans Penman-Monteith equatic and solar radiation. Unlim the required irrigation for
precipitation	Preceding hour sum	mm (inch)	Total precipitation (rain, sl
snowfall	Preceding hour sum	cm (inch)	Snowfall amount of the pr millimeter, divide by 7. E.g
precipitation_probability	Preceding hour probability	%	Probability of precipitatio based on ensemble weath simulations are computed
rain	Preceding hour sum	mm (inch)	Rain from large scale wea
showers	Preceding hour sum	mm (inch)	Showers from convective
weather_code	Instant	WMO code	Weather condition as a nu table below for details.
snow_depth	Instant	meters	Snow depth on the groun
freezing_level_height	Instant	meters	Altitude above sea level o
visibility	Instant	meters	Viewing distance in meter

Variable	Valid time	Unit	Description
soil_temperature_0cm **soil_temperature_6cm** **soil_temperature_18cm** **soil_temperature_54cm**	Instant	°C (°F)	Temperature in the soil at land or water surface tem
soil_moisture_0_to_1cm **soil_moisture_1_to_3cm** **soil_moisture_3_to_9cm** **soil_moisture_9_to_27cm** **soil_moisture_27_to_81cm**	Instant	m^3/m^3	Average soil water conten cm depths.
is_day	Instant	Dimensionless	1 if the current time step

15-Minutely Parameter Definition

The parameter &minutely_15= can be used to get 15-minutely data. This data is based on NOAA HRRR model for North America and DWD ICON-D2 and Météo-France AROME model for Central Europe. If 15-minutely data is requested for other regions data is interpolated from 1-hourly to 15-minutely.

15-minutely data can be requested for other weather variables that are available for hourly data, but will use interpolation.

Variable	Valid time	Unit
temperature_2m	Instant	°C (°F)
relative_humidity_2m	Instant	%
dew_point_2m	Instant	°C (°F)
apparent_temperature	Instant	°C (°F)
shortwave_radiation	Preceding 15 minutes mean	W/m^2
direct_radiation **direct_normal_irradiance**	Preceding 15 minutes mean	W/m^2

Variable	Valid time	Unit
global_tilted_irradiance global_tilted_irradiance_instant	Preceding 15 minutes mean	W/m²
diffuse_radiation	Preceding 15 minutes mean	W/m²
sunshine_duration	Preceding 15 minutes sum	seconds
lightning_potential	Instant	J/kg
precipitation	Preceding 15 minutes sum	mm (inch)
snowfall	Preceding 15 minutes sum	cm (inch)
rain	Preceding 15 minutes sum	mm (inch)
showers	Preceding 15 minutes sum	mm (inch)
snowfall_height	Instant	meters
freezing_level_height	Instant	meters
cape	Instant	J/kg
wind_speed_10m wind_speed_80m	Instant	km/h (mph, m/s, kno
wind_direction_10m wind_direction_80m	Instant	°
wind_gusts_10m	Preceding 15 min max	km/h (mph, m/s, kno
visibility	Instant	meters
weather_code	Instant	WMO code

Pressure Level Variables

Pressure level variables do not have fixed altitudes. Altitude varies with atmospheric pressure. 1000 hPa is roughly between 60 and 160 meters above sea level.

Estimated altitudes are given below. Altitudes are in meters above sea level (not above ground). For precise altitudes, geopotential_height can be used.

Level (hPa)	1000	975	950	925	900	850	800	700	600	500	4
Altitude	110 m	320 m	500 m	800 m	1000 m	1500 m	1900 m	3 km	4.2 km	5.6 km	7 k

Weather Forecast

All pressure levels have valid times of the indicated hour (instant).

Weather Forecast
Historical Forecast
Previous Model Runs
DWD Germany
NOAA U.S.
Météo-France
ECMWF
UK Met Office
KMA Korea
JMA Japan
MeteoSwiss
MET Norway
GEM Canada
BOM Australia
CMA China
KNMI Netherlands
DMI Denmark
ItaliaMeteo
GeoSphere Austria

Variable	Unit	Description
temperature_1000hPa **temperature_975hPa, ...**	°C (°F)	Air temperature at the specified ressu
relative_humidity_1000hPa **relative_humidity_975hPa, ...**	%	Relative humidity at the specifie pres
dew_point_1000hPa **dew_point_975hPa, ...**	°C (°F)	Dew point temperature at the s cifie
cloud_cover_1000hPa **cloud_cover_975hPa, ...**	%	Cloud cover at the specified pre ure l humidity using Sundqvist et al. (989). cover variables.
wind_speed_1000hPa **wind_speed_975hPa, ...**	km/h (mph, m/s, knots)	Wind speed at the specified pre ure l
wind_direction_1000hPa **wind_direction_975hPa, ...**	°	Wind direction at the specified pressu
geopotential_height_1000hPa **geopotential_height_975hPa, ...**	meter	Geopotential height at the spec ed p meter above sea level of each p ssure ground.

Daily Parameter Definition

Aggregations are a simple 24 hour aggregation from hourly values. The paramete &daily= accepts the following values:

Variable	Unit	Description
temperature_2m_max **temperature_2m_mean** **temperature_2m_min**	°C (°F)	Maximum and minimum daily air temp
apparent_temperature_max **apparent_temperature_mean** **apparent_temperature_min**	°C (°F)	Maximum and minimum daily apparent
precipitation_sum	mm	Sum of daily precipitation (including ra
rain_sum	mm	Sum of daily rain
showers_sum	mm	Sum of daily showers
snowfall_sum	cm	Sum of daily snowfall
precipitation_hours	hours	The number of hours with rain
precipitation_probability_max **precipitation_probability_mean** **precipitation_probability_min**	%	Probability of precipitation
weather_code	WMO code	The most severe weather condition on
sunrise **sunset**	iso8601	Sun rise and set times
sunshine_duration	seconds	The number of seconds of sunshine pe irradiance exceeding 120 W/m², followi be less than daylight duration due to d
daylight_duration	seconds	Number of seconds of daylight per day
wind_speed_10m_max **wind_gusts_10m_max**	km/h (mph, m/s, knots)	Maximum wind speed and gusts on a d
wind_direction_10m_dominant	°	Dominant wind direction
shortwave_radiation_sum	MJ/m²	The sum of solar radiation on a given d

Variable	Unit	Description
et0_fao_evapotranspiration	mm	Daily sum of ET_0 Reference Evapotransp
uv_index_max **uv_index_clear_sky_max**	Index	Daily maximum in UV Index starting fro conditions. Please follow the official WM

JSON Return Object

On success a JSON object will be returned.


```
{
    "latitude": 52.52,
    "longitude": 13.419,
    "elevation": 44.812,
    "generationtime_ms": 2.2119,
    "utc_offset_seconds": 0,
    "timezone": "Europe/Berlin",
    "timezone_abbreviation": "CEST",
    "hourly": {
        "time": ["2022-07-01T00:00", "2022-07-01T01:00", "2022-07-0
        "temperature_2m": [13, 12.7, 12.7, 12.5, 12.5, 12.8, 13, 12
    },
    "hourly_units": {
        "temperature_2m": "°C"
    }
}
```


Parameter	Format	Description
latitude, longitude	Floating point	WGS84 of the center of the weather grid-cell which coordinate might be a few kilometres away from th

Parameter	Format	Description
elevation	Floating point	The elevation from a 90 meter digital elevation mo (see parameter cell_selection). Statistical downscal this elevation. This elevation can also be controlled &elevation=nan is specified, all downscaling is dis used.
generationtime_ms	Floating point	Generation time of the weather forecast in millisec monitoring and improvements.
utc_offset_seconds	Integer	Applied timezone offset from the &timezone= pa
timezone **timezone_abbreviation**	String	Timezone identifier (e.g. Europe/Berlin) and abbre
current	Object	For every chosen current weather variable, the data time specifies the moment at which the data is val seconds used for calculating backward-looking sun 900 seconds (15 minutes) means that aggregated r from the previous 15 minutes.
hourly	Object	For each selected weather variable, data will be retu time array will be returned with ISO8601 timestam
hourly_units	Object	For each selected weather variable, the unit will be
daily	Object	For each selected daily weather variable, data will b Additionally a time array will be returned with ISO
daily_units	Object	For each selected daily weather variable, the unit w

Errors

In case an error occurs, for example a URL parameter is not correctly specified, a JSON error object is returned with a HTTP 400 status code.

```
{
    "error": true,
    "reason": "Cannot initialize WeatherVariable from invalid String v
```

```
        tempeture_2m for key hourly"
}
```

Weather variable documentation

WMO Weather interpretation codes (WW)

Code	Description
0	Clear sky
1, 2, 3	Mainly clear, partly cloudy, and overcast
45, 48	Fog and depositing rime fog
51, 53, 55	Drizzle: Light, moderate, and dense intensity
56, 57	Freezing Drizzle: Light and dense intensity
61, 63, 65	Rain: Slight, moderate and heavy intensity
66, 67	Freezing Rain: Light and heavy intensity
71, 73, 75	Snow fall: Slight, moderate, and heavy intensity
77	Snow grains
80, 81, 82	Rain showers: Slight, moderate, and violent
85, 86	Snow showers slight and heavy
95 *	Thunderstorm: Slight or moderate
96, 99 *	Thunderstorm with slight and heavy hail

(*) Thunderstorm forecast with hail is only available in Central Europe

Features
Pricing
About us & Contact
Licence
Terms & Privacy

Weather Forecast API
Historical Weather API
ECMWF API
GFS & HRRR Forecast API
Météo-France API
DWD ICON API
GEM API
JMA API
Met Norway API

Ensemble API
Climate Change API
Marine Weather API
Air Quality API
Geocoding API
Elevation API
Flood API

External

X
Blog
GitHub
Mastodon
Service status and uptime
Model Updates Overview

© 2022-2026 Copyright: Open-Meteo.com